SECTION 7: STANDARD MEDICAL PROCEDURES

7.1 i-gel® (Supraglottic Airway Device) Insertion Principles

NOTE: A SUPRAGLOTTIC AIRWAY DEVICE DOES NOT PROTECT AIRWAYS FROM ASPIRATION

Indications:

- Patient that is unable to maintain a patent airway (GCS ≤ 8 or absence of gag reflex).
- Cardio-respiratory arrest.

Contraindications:

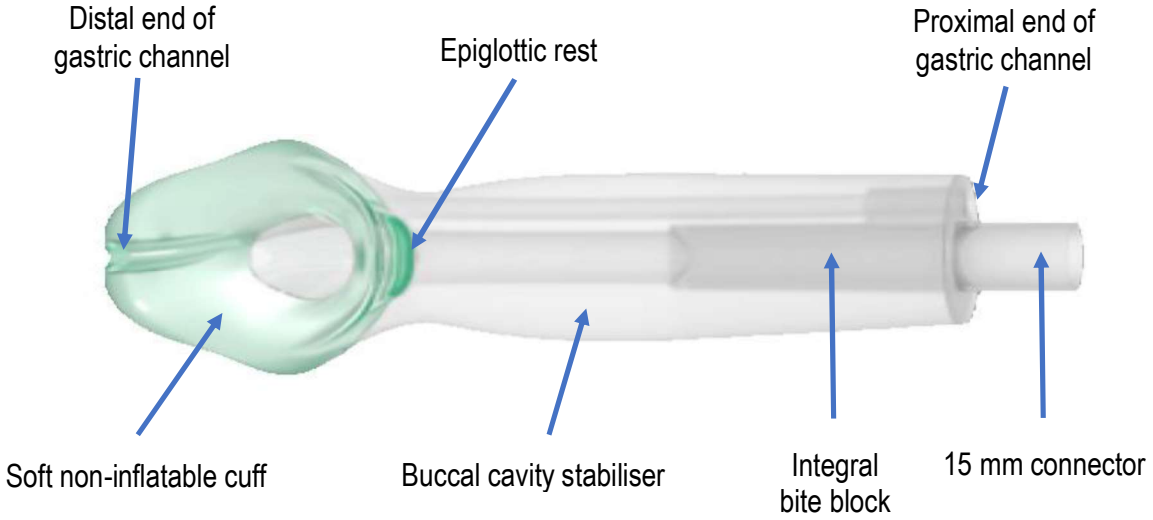
- Gag reflex present.
- Trismus; Limited mouth opening; Pharyngo-perilaryngeal abscess; Trauma or mass.
- Patient with any condition which may increase the risk of a full stomach (e.g.: Hiatus hernia; Sepsis; Morbid obesity; Pregnancy; History of upper gastro-intestinal surgery; etc).

Cautions:

- Do not allow peak airway pressure of ventilation to exceed 40 cm H_2O.
- Do not use excessive force to insert the device.
- Do not leave the device in place for more than four hours.
- Do not reuse or attempt to reprocess the i-gel®.

Key components and their function (i-gel® and Resus Pack)

Description: The i-gel® is made from a medical grade thermoplastic elastomer. i-gel® has been designed to create a non-inflatable, anatomical seal of the pharyngeal, laryngeal and perilaryngeal structures whilst avoiding compression trauma.



Front image of Adult sizes

Front image of Pediatric sizes

SECTION 7: STANDARD MEDICAL PROCEDURES

7.1 i-gel® (Supraglottic Airway Device) continued

Soft non-inflatable cuff:

- Fits snugly onto the perilaryngeal framework, mirroring the shape of the epiglottis, aryepiglottic folds, piriform fossa, peri-thyroid, peri-cricoid, posterior cartilages and spaces.

Gastric channel:

- The gastric channel runs through the device from its proximal opening at the side of the flat connector wing to the distal tip of the non-inflatable cuff. Since the distal tip of the device fits snugly and anatomically correctly into the upper esophageal opening, the distal opening of the gastric channel allows for the passing of a nasogastric tube to empty the stomach contents and can facilitate the venting of gas from the stomach. The gastric channel can also provide an early indication of regurgitation. Please note the size 1 i-gel® does not have a gastric channel.

Epiglottic rest:

- An artificial epiglottis and a protective ridge help prevent the epiglottis from down-folding or obstructing the distal opening of the airway. The epiglottic ridge at the proximal end of the bowl rests at the base of the tongue, thus keeping the device from moving upwards out of position and the tip from moving out of the upper esophagus.

Buccal cavity stabiliser:

- The buccal cavity stabiliser has a built-in natural curvature and an inherent propensity to adapt its shape to the oropharyngeal curvature of the patient. It is anatomically widened and concaved to eliminate the potential for rotation, thereby reducing the risk of malposition. It also provides vertical strength to aid insertion.

15 mm connector:

- Provide a standard 15 mm connection to the anaesthetic system or BVM connection.
- A port of entry for the gastric channel – the port is independent of the main 15 mm connection and is located on the right hand side of the connector wing. Not applicable to size 1 i-gel®.
- An integral bite block – this function is provided by the distal (Below the wing) part of the connector, which runs through the centre of the proximal part of the buccal cavity stabiliser.
- As a guide to correct positioning – the integral part of the bite-block is marked with a horizontally placed black line, which signifies the optimum position of the teeth while the device is in situ (Not applicable to the pediatric sizes).

Types of Presentation:

- i-gel® alone
  - i-gel®
  - Cradle
- i-gel® Resus Pack
  - i-gel®
  - Cradle
  - Airway Support Strap
  - 12FG Suction Tube
  - Water base Lubricant

SECTION 7: STANDARD MEDICAL PROCEDURES

7.1 i-gel® (Supraglottic Airway Device) continued

Assemble the necessary equipment:

1. Select appropriate size airway.

   Color coded full range size:

   - Size 1: Neonate 2-5 kg (4-11 Pounds).
   - Size 1.5: Infant 5-12 kg (11-26 Pounds).
   - Size 2: Small Pediatric 10-25 kg (22-55 Pounds).
   - Size 2.5: Large Pediatric 25-35 kg (55-77 Pounds).
   - Size 3: Small Adult 30-60kg (66-132 Pounds) Resus Pack Yellow.
   - Size 4: Medium Adult 50-90kg (110-198 Pounds) Resus Pack Green.
   - Size 5: Large Adult 90 + kg (Over 198 Pounds) Resus Pack Orange.

2. Water based lubricant (Provided with Resus Pack).
3. Bag Valve Mask (BVM).
4. C-collar (To help stabilizing the i-gel®).
5. Tape or the Airway Support Strap.
6. CO_2 Colorimetric Detector.

Pre-use checks:

- Inspect the packaging and ensure it is not damaged prior to opening.
- Inspect the device carefully and confirm there are no foreign bodies or a bolus of lubricant obstructing the distal opening of the airway or gastric channel.
- Carefully inspect inside the bowl of the device, ensuring surfaces are smooth and intact and also that the gastric channel is patent.
- Discard the device if the airway tubes or the body of the device looks abnormal or deformed.

Pre-insertion preparation:

1. Always wear gloves.
2. Open the i-gel® package and on a flat surface take out the protective cradle containing the device.
3. Place a small bolus of the water-based lubricant (Provided with resus pack) onto the middle of the smooth surface of the cradle in preparation for lubrication. Do not use silicone based lubricant.
4. Grasp the i-gel® along the integral bite block and lubricate the back, sides and front of the cuff with a thin layer of lubricant. This process may be repeated if lubrication is not adequate, but after lubrication has been completed, check that no bolus of lubricant remains in the bowl of the cuff or elsewhere on the device. Avoid touching the cuff of the device with your hands.
5. Place the i-gel® back into the cradle in preparation for insertion.

N.B.: The i-gel® must always be separated from the cradle prior to insertion. The cradle is not an introducer and must never be inserted into the patient's mouth.

SECTION 7: STANDARD MEDICAL PROCEDURES

7.1 i-gel® (Supraglottic Airway Device) continued

Warnings:

- Do not place the device onto a pillow or the patient's chest and always use the protective cradle/cage pack provider.
- Do not use unsterile gauze to help in lubricating the device.
- Do not apply lubricant too long before insertion.

Insertion technique:

1. Grasp the lubricated i-gel® firmly along the integral bite block. Position the device so that the i-gel® cuff outlet is facing towards the chin of the patient.

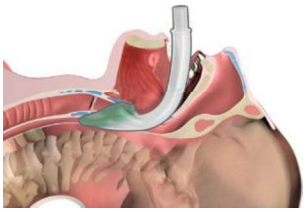
2. The patient should be in ''Sniffing'' position with the head extended and neck flexed (If no spine injury).

3. Introduce the leafing soft tip into the mouth of the patient and direct towards the hard palate.

4. Glide the device downwards and backwards along the hard palate with a continuous but gentle push until a definitive resistance is felt.

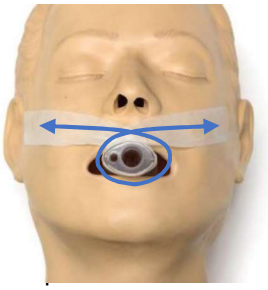
NB: If there is early resistance during insertion, a ''jaw thrust'' is recommended.

5. At this point, the tip of the airway should be located into the upper esophageal opening and the cuff should be located against the laryngeal framework. Once resistance is met and the teeth are located on the bite block, do not repeatedly push i-gel® down or apply excessive force during insertion.



6. i-gel® should be secure with the Airway Support Strap or taped down from ''maxilla to maxilla''.

Airway Support Strap



Taped Down

7.1 i-gel® (Supraglottic Airway Device) continued

7. Attach BVM.

8. Confirm placement:
   a. Auscultate the epigastric region to r/o gastric inflation;
   b. Auscultate breaths sounds bilaterally;
   c. Confirm that the thorax rises evenly;
   d. Attach a CO_2 Colorimetric Detector.

9. Apply C-collar if available.

Important notes:

- Sometimes a feel of ''giving-way'' is felt before the end point resistance is met. This is due to the passage of the bowl of the i-gel® through the faucial pillars (Pharyngo-epiglottic folds).

Incorrect position:

- A horizontal line (Adult sizes 3, 4, 5 only) at the middle of the integral bite-block represents the correct position of the teeth. If the teeth are located lower than the distal tip of the bite block, then it is likely the device has been incompletely inserted. In this instance, remove the i-gel® and reinsert with a gentle jaw thrust applied by an assistant. If that does not resolve the problem, use one size smaller i-gel®.
- The pediatric sizes of i-gel® (Sizes 1 to 2.5) do not have a horizontal line on the integral bite block. This is due to the greater variability in the length of the oro-pharyngeal-laryngeal arch in children. As a result, insertion should continue, as with the adult sizes, until definitive resistance is felt.

Air leakage through the gastric channel:

- A small air leak, or air venting through the gastric channel may be a useful mechanism to protect against gastric insufflation, but an excessive leak means the device is incompletely inserted. In such instances, remove the device and reinsert with a gentle jaw thrust applied by an assistant.

Withdrawal Procedure:

1. Place Casualty in recovery position and have suction ready.

2. Unsecure the tube.

3. Remove the tube.

Monitor patient, record and document procedure

7.11 NPA/OPA Principles

NPA

Indications:

- Patient with a decreased LOC.
- Patient unable to maintain a patent airway.
- NPA can be used prophylactically, if tolerated, in the conscious patient with a patent airway if the provider suspects that the patient's GCS may rapidly deteriorate and/or regular monitoring of the patient/airway may not be possible.

Contraindications:

- Evidence has NOT supported the claim that facial/basilar skull fractures are a contraindication to the placement of an NPA. Correct insertion technique should minimize the risk.
- Patient has no need for an airway adjunct.

Complications:

- Bleeding caused by the insertion may be a complication.
- Mild hemorrhage from the nose after insertion is not an indication to remove it.

NPA Insertion Principles

Measurement:

- The length of the NPA is important. It needs to be long enough to supply an air passage between the patient's tongue and posterior pharynx.
- The distance from the patient's nose (Tip) to the earlobe is a good estimate for the proper size.
- For an adjustable NPA, move the flange to the position that corresponds with the tip of the nose during sizing.

Insertion:

1. Bring the patient's head and neck into a neutral in-line position.

2. Select the nostril that is the largest and least deviated or obstructed.

3. Measure the NPA.

4. Lubricate the distal tip of the NPA with a water-based lubricant, ensuring not to occlude the lumen of the device.

5. Gently pull backward the tip of the nose (''Piggy'' nose).

6. Insertion should follow an anterior-to-posterior plane, along the floor of the nasal cavity.

7. The bevel should be facing the septum:
   a. For Right nostril: Insert the NPA straight back through the nostril following the natural curvature of the airway.
   b. For Left nostril: Turn the NPA upside down (so that the bevel is toward the septum) and insert straight back through the nostril until you reach the posterior pharynx. Rotate the NPA 180 degrees and advance it down the pharynx.

7.11 NPA/OPA Principles continued

8. If resistance is met at the posterior end of the nostril, a gentle back-and-forth rotation of the NPA between the fingers will usually aid in passing it beyond the turbinate bones of the nostril.

9. Should the NPA continue to meet with resistance, the NPA should not be forced past the obstruction but rather withdrawn, and the distal tip should be relubricated and inserted into the other nostril.

10. Continue insertion until the flange end of the NPA is next to the anterior nares or until the patient gags.

11. If patient gags or coughs, it can be a sign that the end of the NPA tube is in contact with the upper part of the larynx and has to be withdrawn slightly.

12. Verify appropriate sizing/placement of the NPA by visually confirming it has reached the pharynx. If sized and inserted correctly, you should not see the distal tip of the NPA as it sits posterior to the base of the tongue.

OPA

Indications:

- Patient who is unable to maintain a patent airway.
- Patients who are easily able to tolerate an OPA should be considered candidates for SGA intubation.

Contraindications:

- Patient who is conscious or semiconscious.
- Patient with an intact gag reflex.

Complications:

- Because it stimulates the gag reflex, use of the OPA may lead to gagging, vomiting, and laryngospasm in patients who are conscious or have an intact gag reflex.
- If the device is too long, it can push the epiglottis over the opening of the larynx obstructing airflow to the trachea and causing a complete obstruction of the airway.
- If not inserted properly, it may push the tongue back into the airway, causing complete or partial obstruction.

OPA Insertion Principles

Measurement:

- The distance from the corner of the mouth to the earlobe (or to the angle of the jaw) is a good estimate for the correct size OPA.

Insertion:

1. Insert the OPA upside down or sideways and rotate into place after tip of OPA passes the tongue. This method should ***not*** be used for children.

2. The flanges of the OPA should be resting against the outside surface of the patient's teeth.

3. If OPA causes gagging, remove it and replace it with NPA.

4. If OPA is displaced, remove it and redo the insertion procedure. ***Do not push it back into place.*** It might push the tongue back, obstructing the airway.

7.3 Transtracheal Block

Indication

- Performing cricothyroidotomy on an awake patient

Contraindication

- Hematoma or burn over the anterior neck

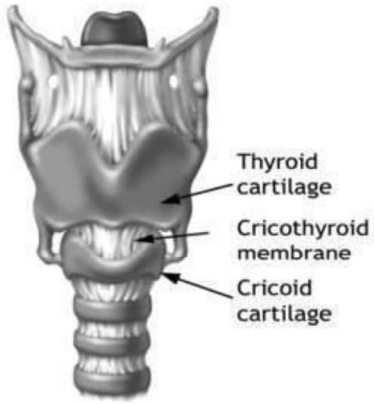
Procedure:

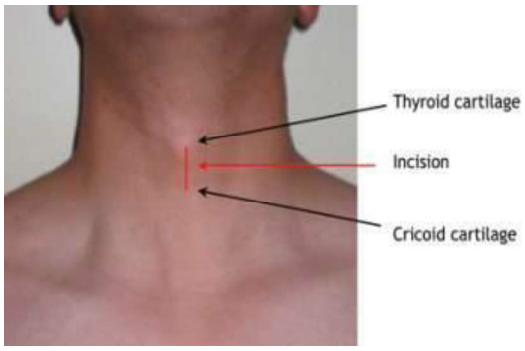
1. Draw up 10 mL 1% Xylocaine
2. Palpate patient's neck and identify cricothyroid membrane
3. Inject 2 mL Xylocaine subcutaneously directly over cricothyroid membrane
4. Inject 2 mL Xylocaine subcutaneously 2 cm cephalad above the cricothyroid membrane
5. Inject 2 mL Xylocaine subcutaneously 2 cm caudad below the cricothyroid membrane
6. Re-landmark and identify cricothyroid membrane
7. At 90 degrees push needle through cricothyroid membrane into trachea
8. Withdraw air into syringe to confirm placement in trachea
9. Rapidly inject remaining 4 mL Xylocaine into trachea and immediately withdraw needle[1]
10. Perform cricothyroidotomy

[1] Expect patient to cough. Though this improves anaesthesia, it will potentially push your needles posterior which might injure the posterior trachea or penetrate the oesophagus.

SECTION 7: STANDARD MEDICAL PROCEDURES

7.4 Cricothyroidotomy Procedure





Indications:

- Airway obstruction due to injuries to the face or neck in which blood or disrupted anatomy precludes the ability to secure an airway by any other means.
- Inhalation burns that compromise the airway.
- Chemical inhalation injury that compromises the airway.
- Anaphylaxis that compromises the airway.

Procedure:

1. If patient is conscious and time/conditions permit, perform a transtracheal block. (Procedure 7.3)

2. Assemble equipment:
   a. Cricothyroidotomy kit;
   b. End-tidal CO_2 Detector;
   c. Bag-Valve-Mask device.

3. Place casualty in supine position.

4. Hyperextend the casualty's neck unless you suspect a Cervical Spine Injury (CSI). Even in a casualty with a suspected CSI, an emergency surgical airway takes precedence. Every effort should be made to avoid movement of a potentially unstable CSI, but hyperextension is appropriate if a cricothyroidotomy cannot be performed effectively and without delay, in a neutral position.

5. Clean the area with iodine and/or alcohol swabs using aseptic technique.

6. Stabilize the larynx between your thumb and middle finger, ensuring not to pull the skin over the larynx to the left or right. Make a vertical incision 2-3 cm (3/4 – 1 inch) long midline over the cricothyroid membrane.

7. Retract the skin around the incision by applying slight downward pressure. Palpate the cricothyroid membrane with your index finger.

8. Lift your index finger and while still maintaining stabilization with your thumb and middle finger, puncture the membrane with the scalpel at 90 degrees to the patient. Extend the incision one scalpel blade width in both directions, to the patient's left and right. The scalpel blade should not be removed from the trachea (Until the tracheal hook is inserted, it is essential to maintain your landmark).

7.4 Cricothyroidotomy Procedure continued

9. Using your non-dominant hand, slide the tracheal hook along the scalpel on the inferior side of the blade until you feel the posterior wall of the trachea and lift upward hooking the trachea.

10. Once trachea is hooked, remove the scalpel.

11. While maintaining tracheal traction, insert tube approximately 7.5 cm (3 inches) into trachea.

12. Inflate balloon.

13. Maintaining position of the tube, attach the BVM. Auscultate the epigastrium to rule out gastric insufflation. Ensure symmetrical chest rise and good breath sounds bilaterally. Confirm effective ventilation with a colorimetric end-tidal CO_2 monitor.

14. Secure tube in place with supplied device.

Monitor and reassess casualty's respirations on a regular basis.